AQA 3.1.4 and 3.1.10 Equilibrium constants K_c and K_p

AQA A-level Chemistry · Revision questions · Sheet 1 of 9

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

Sheet 1 of 9

20 marks · about 25 minutes

Key information

- Mole fraction = moles of that gas ÷ total moles of gas
- Partial pressure = mole fraction × total pressure
- K_p uses partial pressures in round brackets, e.g. $K_p = p(C)^2 \div p(A)\ p(B)^3$; never square brackets
- Find equilibrium amounts with a start / change / equilibrium table using the mole ratio in the equation
- K_c: divide each amount by the volume in dm^3 before substituting (unless the volumes cancel)

Section A: Short-answer questions

01 The equation for the reaction between sulfur dioxide and oxygen is shown. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

In an experiment, 2.00 mol of sulfur dioxide are mixed with 2.00 mol of oxygen. The total amount of the three gases at equilibrium is 3.40 mol.

What is the mole fraction of sulfur trioxide in the equilibrium mixture?

[1 mark]

A 0.176
B 0.353
C 0.600
D 1.200

02 An equilibrium mixture contains 0.40 mol of N_2, 1.20 mol of H_2 and 0.20 mol of NH_3. Calculate the mole fraction of NH_3.

[1 mark]

03 Which equilibrium has a lower yield of product when the pressure in the system is increased at constant temperature?

[1 mark]

A $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$
B $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
C $I_2(g) + H_2(g) \rightleftharpoons 2HI(g)$
D $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

04 Give an expression for K_p for $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$ and deduce its units (pressures in kPa).

[2 marks]

Section B: Longer questions

05 Nitrogen dioxide decomposes at a high temperature.

$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g) \quad \Delta H = +113\ kJ\ mol^{-1}$

05.1 A 0.317 mol sample of nitrogen dioxide is placed in a sealed flask and heated at a constant temperature until equilibrium is reached. At equilibrium, the flask contains 0.120 mol of oxygen. Calculate the mole fraction of each substance at equilibrium.

[3 marks]

05.2 The total pressure in the flask is 120 kPa at equilibrium. Calculate the partial pressure, in kPa, of NO_2. (If you were unable to answer the previous part, assume that the mole fraction of NO_2 is 0.380. This is not the correct answer.)

[1 mark]

05.3 The table shows the mole fractions of the three gases in a different equilibrium mixture.

Gas	Mole fraction
NO_2	0.310
NO	0.460
O_2	0.230

For this equilibrium mixture, K_p = 59.7 kPa

Give an expression for K_p for this reaction. Use your expression and the data in the table to calculate the total pressure, in kPa, in the flask.

[3 marks]

05.4 The equilibrium mixture is compressed into a smaller volume. Deduce the effect, if any, of this change on the equilibrium yield of oxygen and on the value of K_p.

[2 marks]

05.5 The equilibrium mixture is allowed to reach equilibrium at a lower temperature. Explain why the equilibrium yield of oxygen decreases.

[2 marks]

06 Sulfur dioxide reacts with oxygen to form sulfur trioxide.

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

06.1 Give an expression for the equilibrium constant (K_c) for this reaction.

[1 mark]

06.2 A mixture of sulfur dioxide and oxygen is allowed to reach equilibrium in a container of volume 1800 cm^3 at temperature T. At equilibrium, the mixture contains 0.176 mol of sulfur dioxide and 0.461 mol of sulfur trioxide. At temperature T, K_c = 15.0 mol^{-1} dm^3. Calculate the amount, in moles, of oxygen at equilibrium.

[3 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**B**	*Source: 7405/3 Jun 2019 Q11*	1
02	0.20 ÷ 1.80 = 0.111	*Source: Practice question (written for this set)* Allow 0.11 or 1/9	1
03	**A**	*Source: 7404/1 Jun 2025 Q15*	1
04	$K_p = p(CO) \times p(H_2)^3 \div (p(CH_4) \times p(H_2O))$ kPa^2	*Source: Practice question (written for this set)*	2
05.1	M1 NO_2 0.176 M2 NO 0.549 M3 O_2 0.275	*Source: 7405/1 Jun 2023 Q04* Allow answers to 2 sf	3
05.2	21.1 (kPa)	Allow 21.6 kPa Using 0.380: 45.6 kPa	1
05.3	M1 $K_p = p(NO)^2\, p(O_2) \div p(NO_2)^2$ M2 $P = 59.7 \times 0.31^2 \div (0.46^2 \times 0.23)$ M3 = 117.9 or 118 (kPa)	Do not allow square brackets	3
05.4	Yield of oxygen decreases K_p no change		2
05.5	M1 reaction is endothermic / exothermic in backwards direction M2 equilibrium shifts backwards / to the left to raise the temperature / oppose the decrease in temperature		2
06.1	$K_c = [SO_3]^2 \div [SO_2]^2[O_2]$	*Source: 7404/1 Nov 2021 Q10.1–10.2*	1
06.2	M1 dividing by volume: $[SO_2]$ = 0.0978 and $[SO_3]$ = 0.256 mol dm^{-3} M2 $[O_2] = 0.256^2 \div (0.0978^2 \times 15.0)$ (= 0.457 mol dm^{-3}) M3 $n(O_2) = 0.457 \times 1.80 = 0.823$ mol	At least 2 sf	3